

Summary of Diabetes Dataset Analysis

The diabetes dataset was analyzed to understand its structure, identify data-quality issues, and prepare the data for further analysis. The dataset contains demographic and clinical measurements including AGE, Urea, Creatinine (Cr), HbA1c, BMI, and lipid-related variables such as Chol, TG, HDL, LDL, and VLDL.

Data Cleaning and Preparation

The dataset was first examined to identify numerical and categorical variables. The identifier columns were renamed for better clarity, with ID changed to Visit_ID and No_Patien changed to Patient_ID. The categorical variables were also checked for their unique values. Gender contained Female (F) and Male (M), while the diabetes classification contained N (No Diabetes), P (Pre-diabetes), and Y (Diabetes).

Missing values were examined using both counts and percentages. Rows containing missing values were removed where required by the preprocessing procedure. Duplicate and inconsistent records were also considered during data cleaning.

Outlier Analysis

Outlier treatment was performed according to the requirements of each variable rather than applying one method to all columns. Extreme observations in **AGE, HbA1c, and BMI** were retained because these values may represent genuine patient characteristics and removing them could result in loss of meaningful clinical information.

For the **Creatinine (Cr)** variable, observations exceeding the **99.5th percentile** were removed. For **Urea**, observations exceeding the **99.9th percentile** were removed. These specific percentile thresholds were used instead of IQR or other methods.

For the lipid-related variables **LDL, VLDL, HDL, TG, and Chol**, extreme outliers were identified using the **IQR method** and observations outside the calculated acceptable range were removed.

Feature Encoding and Scaling

The categorical Gender variable was label encoded so that Female and Male could be represented numerically for analysis or machine-learning applications.

The numerical measurements were examined for differences in scale. AGE, Urea, Cr, HbA1c, Chol, TG, HDL, LDL, VLDL, and BMI have substantially different ranges. For example, Cr ranges from 6 to 401, whereas HDL ranges from 0.4 to 1.9. Therefore, **StandardScaler** was selected to standardize these variables to a common scale with approximately a mean of 0 and standard deviation of 1.

Identifier columns such as Unnamed: 0, Visit_ID, and Patient_id were excluded from scaling because their numerical values represent identifiers rather than meaningful clinical measurements.

Overall Findings

The analysis shows that the dataset contains clinical variables with substantially different numerical scales and several potential extreme observations. Appropriate preprocessing was therefore necessary before conducting further statistical analysis or building machine-learning models. The combination of missing-value handling, selective outlier treatment, categorical encoding, and feature standardization provides a cleaner and more consistent dataset while preserving clinically meaningful observations where appropriate.

Overall, the preprocessing steps improve the reliability of subsequent analysis and ensure that variables with larger numerical ranges do not disproportionately influence statistical or machine-learning methods.